HOPPER INFORMATION SERIES

03 / BLOCK CODING

Coding blocks reference

What each Hopper Studio block category does, what it returns and how commands sequence.

FLIGHT

VISION

LOGIC + LOOPS

EXECUTION MODEL

How a block program becomes a flight

1 Arrange
Statement blocks form ordered stacks.

→

2 Generate
Blockly creates asynchronous JavaScript.

→

3 Run
Commands execute in sequence with await.

→

4 Observe
Active action blocks glow while running.

→

5 Finish
Normal main-program completion auto-lands.

- **when program starts**
  - **take off**
  - **fly forward for 1 sec at 15%**
  - **land**

GENERATED JAVASCRIPT

```
await runtime.runBlock("takeoff", async () => {
  await drone.takeOff();
});
await runtime.runBlock("fly-1", async () => {
  await drone.fly("forward", 1, 15);
});
await runtime.runBlock("land", async () => {
  await drone.land();
});
```

GENERAL CONTROL

Start & Events

when program starts

Container for the main stack. Other top-level stacks can also be generated.

when [key] is [pressed/released]

Arrow, Space and a-z events. Event programs keep listening until Stop.

stop program

Ends execution but does NOT send a land command. Land first or use UI STOP & LAND.

[key] key is pressed

Boolean from the live pressed-key set.

wait [seconds]

Pauses in sequence. Negative/coercion failures become zero.

continue if [condition]

False returns from the current main, event or function.

print [value]

Writes a line to the in-app program console.

SAFETY

The red UI STOP & LAND is the dependable way to cancel tasks and send a landing command.

MOTION COMMANDS

Mini Drone: flight blocks

take off / land / hover

Awaited state-changing actions.

fly [direction] for [s] at [power]%

6 directions; time >= 0; teach 0-100% power; default 1 s / 15%.

rotate [degrees] [direction]

Timed/open-loop yaw at about 180 deg/s, then settle.

flip [direction]

Forward/back/left/right; preserve safe height and clear space.

set [axis] to [power]%

Persistent pitch/roll/yaw/gaz power until reset. 'Altitude' means gaz, not height.

reset movement

Zeros all motion axes; does not land.

center on AprilTag

Repeated scans, translation pulses and yaw alignment; does not control height or land.

cut off motors

Emergency only. Immediate; block/code has no confirmation.

Fly adds a stabilization wait after motion. set-axis does not.

LIMITED EXPOSED TELEMETRY

Mini Drone: state, events and accessories

SENSORS + EVENTS

- battery level -> number 0-100 or null before telemetry.
- drone is flying / landed -> Boolean.
- wait until battery changes -> asynchronous wait.
- when drone starts flying / lands / crashes / battery changes.

ACCESSORIES

- take and store photo -> session gallery JPEG from camera pixels.
- open / close grabber -> needs physical claw; simulator is a timing stub.
- fire cannon -> needs physical cannon; simulator is a timing stub.

NOT AVAILABLE AS BLOCK VALUES

GPS | x/y room position | real altitude | velocity | attitude | heading | raw IMU | range

'set altitude' is signed vertical motor power. It is not an altitude setpoint.

EVERY PREDICATE CAN TRIGGER REAL COMPUTATION

Camera Vision blocks

BINARY

- sees color: threshold + coverage
- center pixel color: threshold

OBJECTS

- scan objects (default 55%)
- sees label at confidence %
- last label x/y at confidence %

CUSTOM

- sees custom label at confidence %

APRILTAGS

- scan AprilTags
- sees tag ID any / 0-586
- center tag: power + slack + lost

Binary: threshold + coverage are 0-100%; invert swaps classes; predicates await a fresh frame and return Boolean.

COCO: label + confidence + saved coordinate (-100..100, or 0). Custom: label + confidence only; no box or coordinate.

AprilTags: scan returns detections; sees/center return Boolean; center also stops at 30 s or the lost-search limit.

BUILT-IN BLOCKLY TOOLS

Logic, loops, math, variables and functions

LOGIC

- if / else
- comparisons
- and / or / not
- true / false
- ternary

LOOPS

- forever
- for N seconds
- repeat count
- while / until
- counted for
- break / continue

MATH

- arithmetic
- unary math
- trig
- rounding
- modulo
- random

VARIABLES

- create
- get
- set
- change

FUNCTIONS

- define
- parameters
- return value
- awaited calls

Loops yield so Stop can interrupt. Generated custom functions are async; calls are awaited.

RECOMMENDED PATTERN

Build one intentional, safe mission

- when program starts
  - take off
  - set found to false
  - repeat for 10 seconds
    - if camera sees white paper
      - set found to true
      - break out of loop
    - else: fly forward 0.5 sec at 12%
  - land after the loop

MISSION RULES

- Place all intended executable stacks deliberately; detached top-level stacks may still run.
- Use one start hat for the main mission.
- Break after the target is found; keep land outside the loop so every normal path lands.
- Use the UI STOP & LAND for intervention.

SOURCES AND FURTHER READING

Hopper Studio block definitions and JavaScript generators
lib/blockly.ts in the Hopper Studio project

Hopper Studio execution and safety behavior
components/HopperStudio.tsx, lib/runtime.ts and README.md